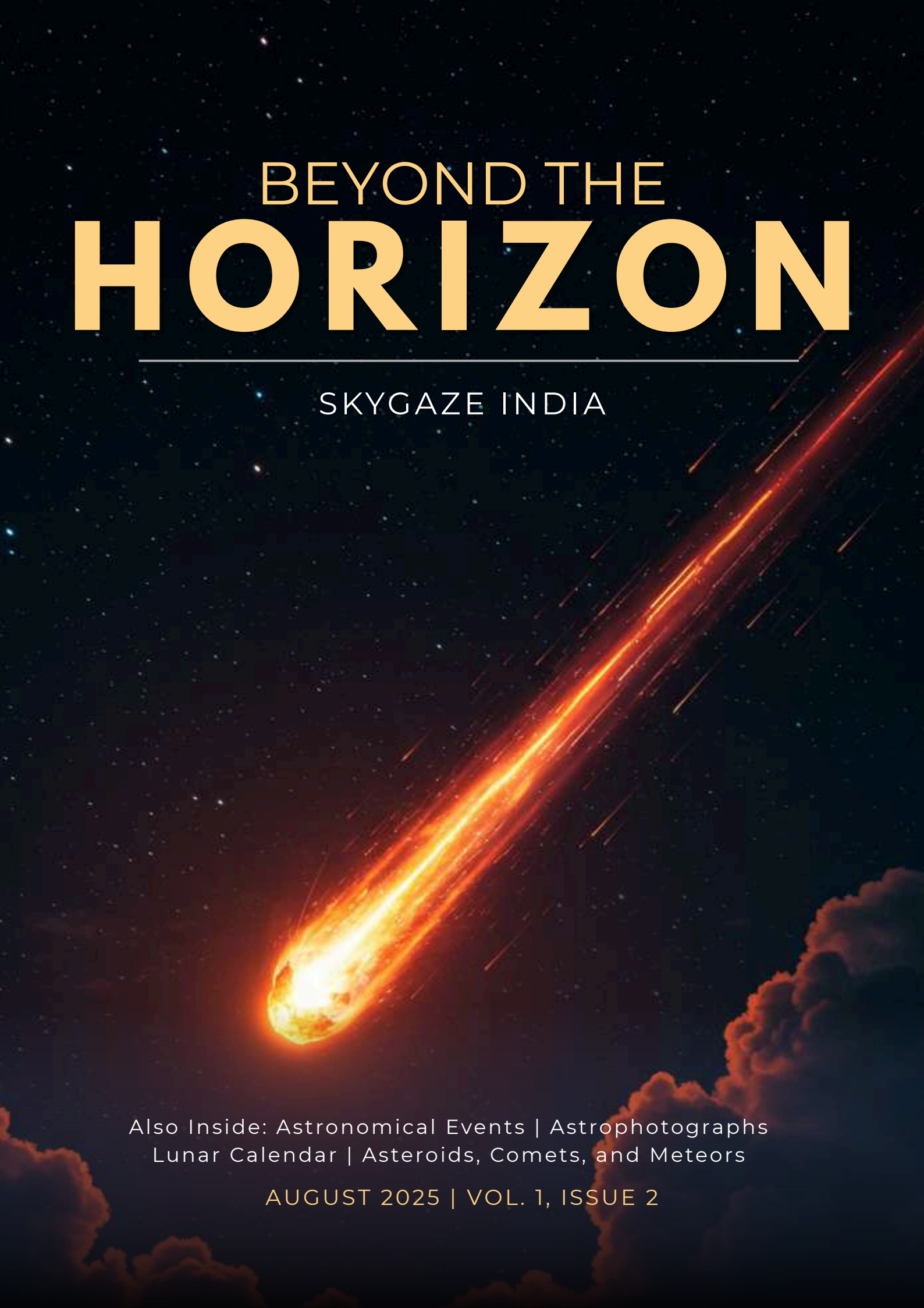




BEYOND THE HORIZON

SKYGAZE INDIA

Also Inside: Astronomical Events | Astrophotographs
Lunar Calendar | Asteroids, Comets, and Meteors

AUGUST 2025 | VOL. 1, ISSUE 2

INTRODUCTION

Beyond the Horizon, Skygaze India's monthly publication, is meticulously curated to stimulate intellectual curiosity and foster deeper public engagement with astronomy and space science.

Each issue presents an amalgamation of celestial event highlights, exemplary astrophotography features, perspicacious articles, and exclusive content contributed by our community of astronomy enthusiasts. Our objective is to make space science both accessible and profoundly inspiring.

OUR TEAM

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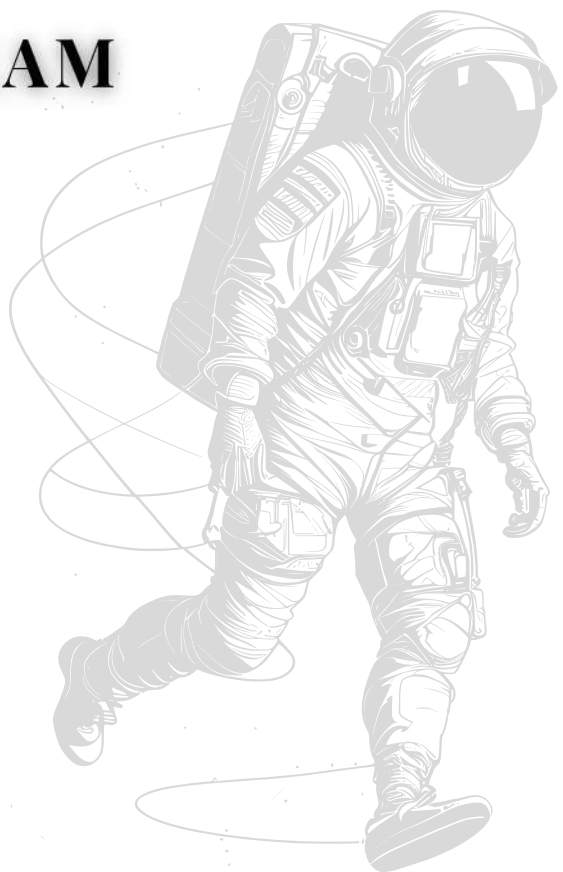
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"At Skygaze India, our mission is to make space science accessible—to bring the cosmos closer by turning complex knowledge into everyday curiosity for all."

Harsh
Founder & CEO

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Discover the celestial events visible from India

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Build a tool to determine which stars and constellations are visible on any given night

ASTRONOMICAL EVENTS



AUGUST 9: STURGEON FULL MOON

Named after the sturgeon fish, this luminous full moon will ascend shortly after sunset. Easily visible across India with the naked eye, it's perfect for moon photography or for viewing lunar features using a telescope or binoculars.

AUGUST 12 – CLOSE CONJUNCTION OF VENUS & JUPITER

Venus and Jupiter will appear remarkably close in the eastern pre-dawn sky. Observers should look for these two brilliantly shining celestial bodies side-by-side approximately one hour before sunrise.



AUGUST 12 & 13 – PERSEIDS METEOR SHOWER

One of the year's most anticipated astronomical phenomena will reach its peak activity. Under dark skies after midnight, a display of 60–100 meteors per hour is anticipated. No equipment needed- simply lie back and enjoy.

AUGUST 23: NEW MOON

The absence of the Moon will offer ideal conditions for deep-sky observations. Indian stargazers can enjoy clearer views of distant galaxies and nebulae- especially from dark, rural areas away from city lights.



AUGUST 27: MERCURY AT PERIHELION

Mercury will be at its closest point to the Sun, moving swiftly across the sky. Although tricky to spot, it might be visible just after sunset in western sky. For the best chance of observation, use binoculars and ensure a clear horizon.

LUNAR CALENDAR

Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
					01  52.41% First Quarter 7.61 day	02  61.78% First Quarter 8.5 day
03  70.8% Waxing Gibbous 9.4 day	04  79.17% Waxing Gibbous 10.31 day	05  86.58% Waxing Gibbous 11.24 day	06  92.7% Waxing Gibbous 12.19 day	07  97.16% Full Moon 13.17 day	08  99.61% Full Moon 14.18 day	09  99.78% Full Moon 15.21 day
10  97.48% Full Moon 16.26 day	11  92.71% Waning Gibbous 17.34 day	12  85.66% Waning Gibbous 18.42 day	13  76.69% Waning Gibbous 19.5 day	14  66.3% Last Quarter 20.59 day	15  55.08% Last Quarter 21.67 day	16  43.64% Last Quarter 22.75 day
17  32.58% Last Quarter 23.82 day	18  22.47% Waning Crescent 24.89 day	19  13.84% Waning Crescent 25.95 day	20  7.1% Waning Crescent 27 day	21  2.53% New Moon 28.03 day	22  0.27% New Moon 29.04 day	23  0.29% New Moon 0.51 day
24  2.45% New Moon 1.48 day	25  6.52% Waxing Crescent 2.43 day	26  12.22% Waxing Crescent 3.36 day	27  19.24% Waxing Crescent 4.27 day	28  27.29% Waxing Crescent 5.17 day	29  36.1% First Quarter 6.06 day	30  45.4% First Quarter 6.95 day
31  54.91% First Quarter 7.85 day						



MIND-BLOWING FACT OF THE MONTH

The Moon is slowly drifting away from Earth—by about 1.5 inches (3.8 cm) per year!
 When it first formed 4.5 billion years ago, it was 15 times closer to Earth—making it look much larger and causing more extreme tides!

MONTH RECAP

SKYGAZE INDIA AT BIOENERGY EXPO GLOBAL 2025

We are proud to share that Skygaze India successfully collaborated with NCBI for the prestigious BioEnergy Global Expo 2025 held from 29th to 31st July at Yashobhoomi, Delhi, and powered by the Ministry of New and Renewable Energy.

Over the three-day event, thousands of delegates, visitors, and organizers experienced the wonders of the Sun through high-powered solar telescopes. Beyond safe solar observation and insightful discussions, our setup became a hub of curiosity, learning, and valuable networking opportunities with other attendees.

This event provided a valuable opportunity to showcase astronomy's relevance at such a forward-looking expo focused on sustainability and innovation.



COLLABORATION WITH CASSIOPEIA, EGYPT

In July, Skygaze India proudly established a collaborative partnership with Cassiopeia in a shared mission: to enhance reach and understanding of space science. This cross-border collaboration marks a new chapter in global astronomy outreach—uniting passion, education, and creativity.

Together, we're set to launch educational initiatives, produce diverse cross-cultural space content, and inspire the next generation of sky enthusiasts. This partnership unequivocally reflects the profound belief that space serves as a universal language—and now, we're speaking it louder than ever.



Mohamed Omar Zeyada — From Desert to Space

We extend our heartfelt congratulations to Mohamed Omar Zeyada, the founder of Cassiopeia, on his selection as Egypt's third Analog Astronaut for the prestigious MMAARS Analog Astronaut Training Program.

Raised in a small village in Fayoum, Egypt, Mohamed's journey to the stars began with limited resources but unlimited determination. He diligently pursued certifications in telecommunications, computer engineering, astronomy, and space science—proving that passion can outshine any circumstance.

Through Cassiopeia, Mohamed and his dedicated team have significantly contributed to international publications like *Beyond the Horizon*, developed astronomical imaging resources, and advanced scientific outreach under the powerful motto: #FromDesertToSpace.

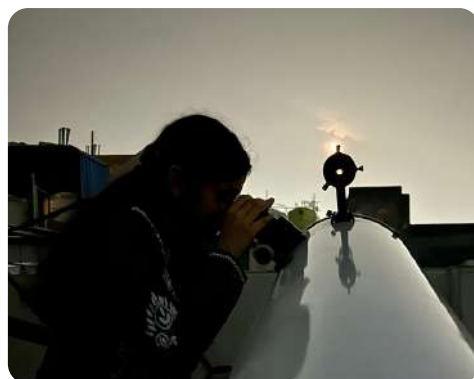


STARGAZING HIGHLIGHTS

"We had a really great experience at Skygaze India, we loved it. We saw the moon in detail and also saw the stars Arcturus and Vega."



Amit
Stargazing Enthusiast



ASTROPHOTOGRAPHS



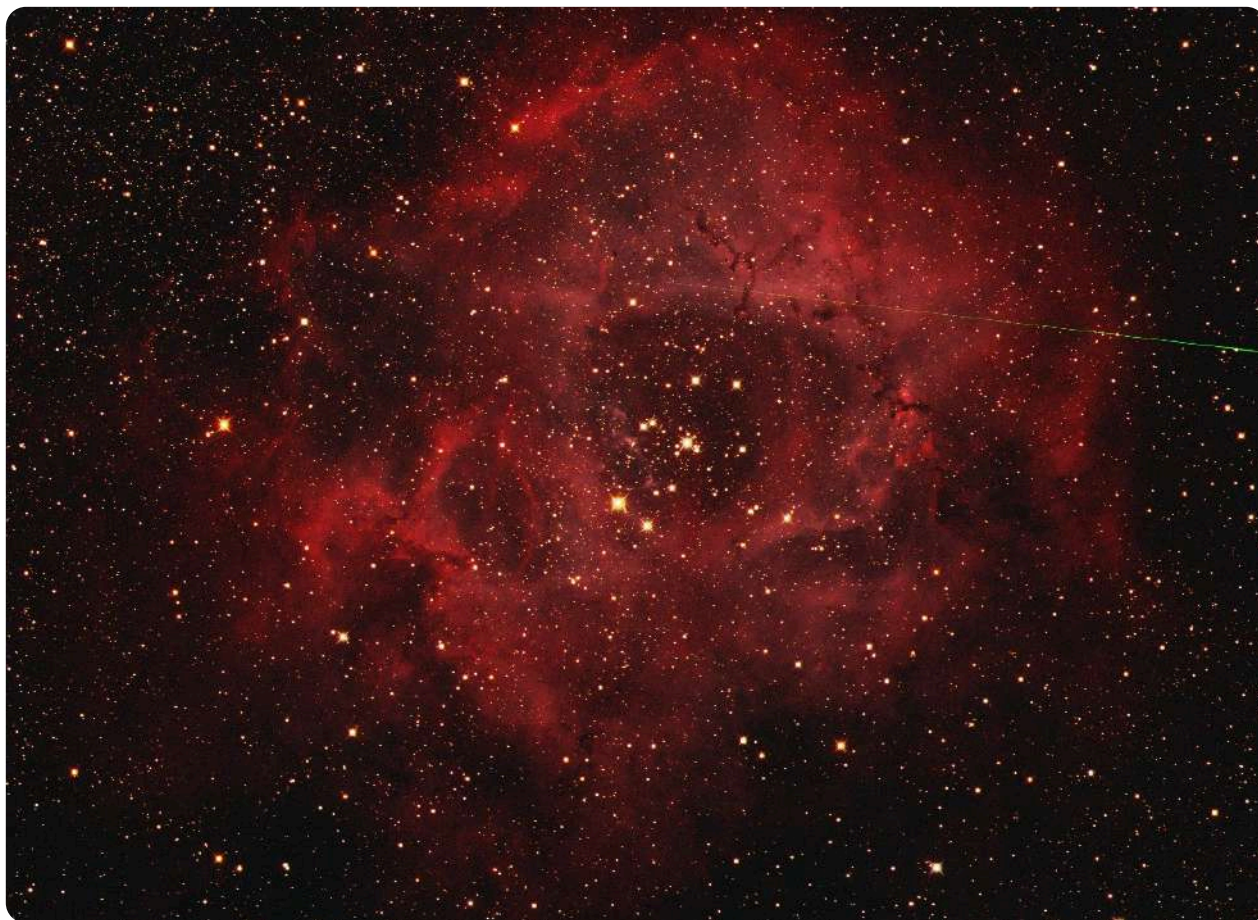
Whirlpool Galaxy captured by Mr. Alaa Aljabo, Member of Team Paradox, Egypt.



Trifid Nebula captured by Mr. Alaa Aljabo, Member of Team Paradox, Egypt.



Horsehead Nebula captured by Mr. Hams Mohamed, Member of Team Paradox, Egypt.



Rosette Nebula captured by Mr. Mohamed Elattar, Supervisor of Team Paradox, Egypt.



Buck Full Moon, captured by Mr. Harsh, Founder at Skygaze India, Delhi, India

FEATURE YOUR WORK NEXT MONTH !

Have an astrophotograph you'd love to share with the world? We're inviting submissions for our next issue of Beyond the Horizon, a Skygaze India publication.

Submit via email on
support@skygazeindia.com



UNDERSTANDING ASTEROIDS, COMETS, AND METEORS

Have you ever wondered what the difference is between an asteroid, a comet, or a shooting star? While they may seem alike, each has a unique identity and origin in space.

Asteroids – The Rocky Relics

Asteroids are rocky, airless bodies that orbit the Sun, mostly found in the **Asteroid Belt** between Mars and Jupiter. They are considered leftovers from the early formation of our solar system.

They range in size from tiny pebbles to massive rocks over 900 km wide (like Ceres). Most are irregularly shaped due to low gravity.

NASA keeps a close watch on Near-Earth Asteroids (NEAs) — those that come close to Earth's orbit — to ensure our planet stays safe from potential impacts.



Comets – The Icy Wanderers

Comets are made of **ice, dust, and rocky material**. They originate in the cold outer regions of the solar system — mainly the **Kuiper Belt** and **Oort Cloud**.

As a comet approaches the Sun, it heats up, and the ice begins to vaporize, creating a glowing **coma** (cloud around the nucleus) and often a spectacular **tail** that points away from the Sun due to solar wind.

Halley's Comet, which is visible from Earth every 76 years, is perhaps the most well-known.

Meteor Showers

Events like the Perseids and Leonids occur when Earth passes through the debris trail of a comet, producing numerous meteors in the sky. Studying these space objects gives scientists clues about how our solar system formed — including the origins of planets, water, and possibly even life. Some missions, like NASA's OSIRIS-REx, have collected samples from asteroids to study their composition.

Whether it's a distant icy comet, a rocky asteroid belt traveler, or a meteor lighting up the night sky, these space rocks are not just debris — they are storytellers from the past.





Meteoroids, Meteors, and Meteorites – A Cosmic Chain

These three terms are closely related but describe different stages of the same object:

- **Meteoroid** - A small rock or particle from an asteroid or comet, floating in space.
- **Meteor** - When a meteoroid enters Earth's atmosphere and burns up due to friction, it produces a streak of light — a “shooting star.”
- **Meteorite** - If a meteor survives the journey and lands on Earth, it is called a meteorite.

Fireball - A Bright Meteor Phenomenon

A fireball is a type of meteor that shines exceptionally bright as it travels through Earth's atmosphere. It is significantly brighter than the planet Venus, which is often one of the brightest objects in the night sky after the Moon. Due to their brightness, fireballs can often be seen during the daytime as well, unlike regular meteors.

Bolide - An Exploding Fireball

A bolide is a specific type of fireball that explodes or bursts in the atmosphere with a brilliant flash and often a sonic boom. The term is sometimes used interchangeably with fireball, but in astronomy, a bolide specifically refers to a fireball that disintegrates violently, sometimes causing damage or leaving behind meteorites.

CONTRIBUTE TO BEYOND THE HORIZON!

Have an astronomy article, story, or thought to share?
We're inviting written submissions for our next publication!

Submit via email on support@skygazeindia.com



THIS MONTH IN INDIAN SPACE HISTORY

AUGUST, 1991

BIRTH OF VIKRAM SARABHAI

12

Vikram Sarabhai, often regarded as the father of India's space program, was born on this date. His vision and leadership were instrumental in establishing ISRO in August 1969. Sarabhai's legacy continues to guide India's space ambitions.

AUGUST, 1975

LAUNCH OF SITE

01

The Satellite Instructional Television Experiment (SITE) began on 1 August 1975, broadcasting educational programming to over 2,400 villages across six Indian states. Conducted jointly by NASA and ISRO, SITE was a pioneering application of satellite technology for social upliftment and capacity-building in rural India.

AUGUST, 2023

CHANDRAYAAN-3 LUNAR LANDING

23

India's Vikram lander and Pragyan rover made history with a safe touchdown near the Moon's south pole on 23 August 2023. With this mission, India became the first country to land at the lunar south pole and the fourth nation overall to accomplish a soft moon landing.

**LAUNCHING
SOON**



SKYGAZE ACADEMY

At Skygaze Academy, we're making science simple. As a next-generation educational platform, powered by Skygaze India, we harness the power of AI and IT to demystify complex subjects, with a special focus on space science. Join Skygaze Academy and transform your curiosity into understanding.

EXPLORING THE NIGHT SKY USING A PLANISPHERE

Learning Outcomes:

- Understand how the night sky changes with time
- Identify major constellations and bright stars in the night sky
- Use a planisphere as a tool for celestial navigation
- Recognize the concept of the rotating sky and Earth's position in space

Items Required:

- Printable Planisphere
- Scissors
- Split pin or paper fastener (to assemble rotating disk)
- Compass (to identify directions)
- Clear outdoor space with visible night sky

Instructions:

1. Assemble the Planisphere: Cut out both circular parts and attach the rotating disk using a split pin through the center. Ensure it rotates smoothly.
2. Set Your Date & Time: Rotate the top disk to align your local stargazing time with the current date printed on the wheel.
3. Face North: Go outside to a safe, open space. Use a compass to face North and hold the planisphere above your head, so the N label on the disk faces actual North.
4. Match the Sky: What you see on the planisphere window should now reflect the visible stars and constellations overhead at that time and date. (Try spotting Orion, Ursa Major, Cassiopeia, etc.)
5. Optional – Track Sky Movement: Try using the planisphere again a few hours later or on a different date and observe how the sky “moves” or shifts. Discuss why this happens (Earth’s rotation and orbit).

What is Planisphere?

Planisphere is a star chart analog computing instrument in the form of two adjustable disks that rotate on a common pivot. It can be adjusted to display the visible stars for any time and date. It is an instrument to assist in learning how to recognize stars and constellations.



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